République Tunisienne
Ministère de l'Enseignement Supérieur
et de la Recherche Scientifique

Université de Carthage

Institut Supérieur des Technologies de
l'Information et de la Communication

Internship Report

Bachelor's Degree in Computer Science

Specialization: Software Engineering and Information Systems

Admin Dashboard

By **Ahmed Akrout**

Completed at *Technozor*

Professional Supervisor: Slim Boufaden

Internship Period: 02/07/2024 - 30/08/2024

Academic Year: 2023 - 2024

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Autorisation de dépôt du rapport de Stage :

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General Introduction

Farmalabs is an efficient and collaborative platform that is set up to help administrators manage apiary with ease while optimizing workflows. That is, this web application provides a comprehensive user experience with powerful features to simplify all workflows, such as scheduling, editing, and tracking treatments and inspections, tasks, and user management in one central system.

Farmalabs presents an easy-to-use solution for apiary management: for administrators, it allows them to make follow-ups on treatments and inspections, which also include the weather condition, with the facility to focus deeply on each apiary's information by location. This makes the processes clear while allowing access to value data that enhances productivity at a daily operational level.

Applying best practices in farm management, Farmalabs is positioned as a modern, intuitive tool that enhances operational efficiency. The flexible dashboards and user-friendly interface of the tool enable effective monitoring and management of every aspect of apiary operations with real-time access to data.

This report describes the major characteristics and benefits that Farmalabs offers as an innovative collaboration platform for apiary management. Based on functionality described at the official platform, complemented by other resources, this report explains how Farmalabs helps to improve farm management practices and increases overall productivity.

Chapter 1

Project Context

1.1 Introduction

Technozor is an AI-driven company founded in 2021, focusing on providing Enterprise AI solutions to tackle complex business problems. Their expertise spans data analytics, anomaly detection, computer vision, and natural language processing.

1.2 Presentation of the Host Organization

1.2.1 Host Organization

Figure 1.1: Technozor

1.2.2 Areas of Activity

Technozor assists organizations in designing data strategies, managing data governance, implementing MLOps, and developing custom analytical use cases. Their services aim to optimize processes, improve decision-making, and provide a competitive edge through advanced technologies.

1.3 Project Presentation

1.3.1 Problem Statement

Apiary management is mostly labor-intensive, and there is a headache without any one organized system. Admins go through lots of hassle in effectively scheduling treatments, managing inspections, and tracking tasks among several apiaries. Real-time data and weather updates, when not available in the most handy manner, make decision-making even tougher.

1.3.2 Existing Study

Various researchers have identified different agriculture management systems, showing where technology fits to ensure ease in most of the operations. Most of the identified solutions lack specific features targeted for the apiary, such as scheduling treatments and weather conditions.

1.3.3 Proposed Solution

Farmalabs is intended to make beehive management that easy. Its contribution will be the following:

- Treatment Management: Admin can add or delete treatments in the hive and also keep track of it.
- Manage Inspection: Admins can add, change, or delete inspection records to check hive health.
- Task Tracking: Administrators can view and manage the tasks of every hive.
- Location View: Admins can view information about hives depending on the location.
- Weather Updates: Admins will view real-time weather updates to enable them to make good decisions on behalf of the hives.

1.4 Conclusion

In this chapter, i introduced the hosting organization, Technozor; explained real problems in apiary management derived into an ideal solution which would be able to solve them; and underlined the methodology of work for the success of the Farmalabs project.

Chapter 2

Requirements Specification

2.1 Introduction

In this chapter, i define the functional and non-functional requirements of our system. i also identify the involved actors and present the general architecture through use case diagrams.

2.2 Identification of Functional Requirements

A functional requirement describes what a product or service is supposed to do, i.e., its main functions. These are concrete specifications detailing the operations, features, or tasks that the product or service must be able to execute.

Our Farmalabs system allows the following functionalities:

- Login: The admin must be able to securely log in to the system.
- Manage Inspections: The admin can view the list of inspections, add, edit, and delete inspection entries.
- Manage Tasks: The admin can view the list of tasks, add, edit, and delete tasks.
- Manage Treatments: The admin can view the list of treatments, add, edit, and delete treatments.
- View Information: The admin can access various information related to inspections, tasks, and treatments, including apiary locations, treatment schedules, and more.

2.3 Identification of Non-Functional Requirements

- **Performance:** Performance: Guarantee a fast system response, even under heavy load.
- **Maintainability:** Simplify the maintenance and management of the system by providing efficient tools and processes for updating content and integrating new features as needed.
- **Usability:** Usability: Provide a user-friendly and intuitive interface.
- **Auditability:** Availability: Guarantee system access 24 hours a day, 7 days a week.

2.4 Identification of Actors

Admin: The admin is the sole user of the system with special privileges to manage and oversee all activities and functionalities. The admin has full access to the following features:

Actor	**Roles**
Admin	• Manage treatments (add, edit, delete) • Manage tasks (create, assign, edit, delete) • Manage inspections (add, edit, delete) • View detailed apiary and weather information

Table 2.1: Actor and their roles in the Farmalabs system

2.5 Global Use Case Diagram

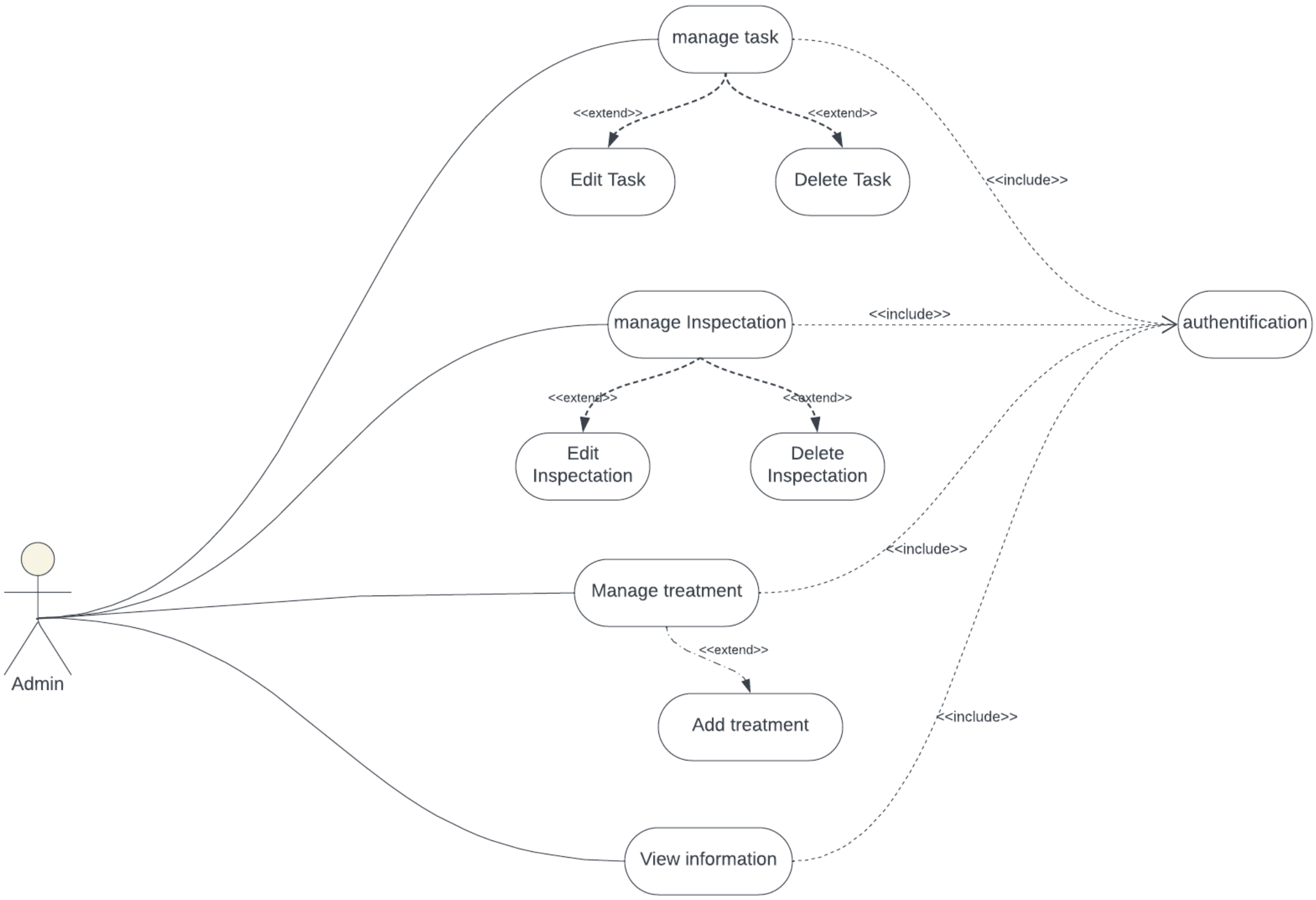

Figure 2.1: Use Case Diagram

2.6 Product Backlog

Product Backlog	Priority	Estimation	Sprint
As an Admin, i can authenticate	1	Medium	sprint 0
As an Admin, i can view information	1	High	sprint 0
As an Admin, i can manage treatments	2	Medium	sprint 1
As an Admin, i can manage tasks	2	Medium	sprint 1
As a Admin, i can manage inspectation	2	Medium	sprint 1

Table 2.2: Backlog table

2.7 Working Environment

2.7.1 Design Methodology

The design methodology in Scrum relies on an iterative and collaborative approach to create products and features within the Scrum framework. Unlike traditional methodologies, where design is often done sequentially upfront, Scrum integrates design into the agile development process. In Scrum, design is a continuous effort that evolves over the course of sprint iterations. This incremental approach aims to optimize predictability and control risks throughout the development cycle.

2.7.2 Software Environment

Figure 2.2: Lucidchart

Lucidchart is a web-based diagramming tool used by people in diverse industries to create detailed diagrams out of images, data, designs, or templates. It allows collaboration with other users and comes with multiple integrations, including Google Workspace and Atlassian, among others to streamline your process.[1]

Figure 2.3: ReactJs

React, also known as ReactJS, is a popular and powerful JavaScript library used for building dynamic and interactive user interfaces, primarily for single-page applications (SPAs). It was developed and maintained by Facebook and has gained significant popularity due to its efficient rendering techniques, reusable components, and active community support.[2]

Figure 2.4: FastApi

FastAPI is a modern, fast (high-performance), web framework for building APIs with Python based on standard Python type hints.[3]

Figure 2.5: Bootstrap

Bootstrap is a powerful front-end framework designed to create responsive and mobile-first websites efficiently. It combines HTML, CSS, and JavaScript, offering a range of components and utilities to streamline web development.[4]

Figure 2.6: CSS

Cascading Style Sheets, is a style sheet language that allows the visual formatting of web pages. Alongside HTML and JavaScript, CSS is one of the core building blocks of the World Wide Web.[5]

Figure 2.7: Thunder Client

Thunder Client is a lightweight REST API Client Extension for Visual Studio Code, hand-crafted by Ranga Vadhineni with a focus on simplicity, clean design, and local storage.[6]

Figure 2.8: GitHub

GitHub is a web-based interface that uses Git, the open source version control software that lets multiple people make separate changes to web pages at the same time. As Carpenter notes, because it allows for real-time collaboration, GitHub encourages teams to work together to build and edit their site content.[7]

Figure 2.9: Visual Studio Code

Visual Studio Code is a streamlined code editor with support for development operations like debugging, task running, and version control. It aims to provide just the tools a developer needs for a quick code-build-debug cycle and leaves more complex workflows to fuller-featured IDEs, such as Visual Studio IDE.[8]

2.8 Conclusion

This chapter provides an overview of the context of our project, starting with a presentation of the host organization. Then, i have defined the general concept of our website, identified the issues, and critiqued existing solutions in order to propose a more innovative approach.

Chapter 3

Sprint 0

3.1 Introduction

This section provides an overview of Sprint 0, focusing on the setup and initial planning phase for the Farmalabs admin dashboard. The goal of Sprint 0 is to establish the foundational elements for future sprints, including system design, backlog identification, and key use cases.

3.2 Identification of Sprint 0 Backlog

Product Backlog	Priority	Estimation	Sprint
As an Admin, i can authenticate	1	Medium	sprint 0
As an Admin, i can view information	1	High	sprint 0

Table 3.1: Sprint 0 backlog

3.3 Sprint 0 Refinement

In this section, we explore the following use cases:

- Authenticate
- View Information

3.3.1 Refinement of the use case "Authenticate"

Figure 3.1: Use Case "Authenticate"

Use case	– Authenticate
Actor	– Admin
Pre-Conditions	– System running – Internet connection – Access to the application
Post-Conditions	– Admin logged in and able to use the application
Main Scenario	– The system displays the authentication interface – The admin inputs his username and password – The Admin presses the login button – The system verifies the username and password – The system displays the dashboard
Exception	– The system displays an error message if the inputs are invalid

Table 3.2: Authenticate refinement table

3.3.2 Refinement of the use case ”View information”

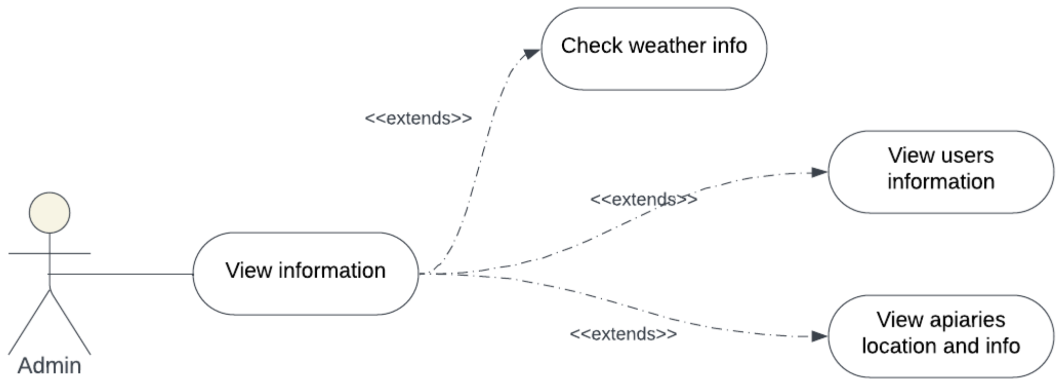

Figure 3.2: Use Case ”View information”

Use case	– View information
Actor	– Admin
Pre-Conditions	– System running – Internet connection – Admin is logged in
Post-Conditions	– Admin successfully views weather information, apiary locations and details,and user contact information, with the option to navigate to other sections.
Main Scenario	– Admin is presented with weather information as the default view. – Admin can navigate to other sections to view more information: – Apiaries Page – The system displays a list of apiaries with their locations and details. – Contact Page: – The system displays user information.

Table 3.3: View information refinement table

3.4 Sprint 0 Conception

3.4.1 conception of the use case "Authenticate"

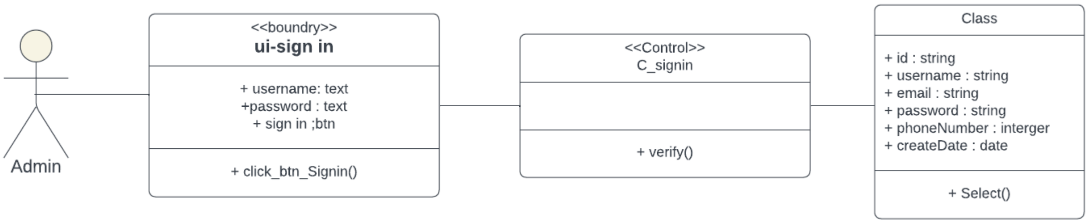

Figure 3.3: Class Diagram "Authenticate"

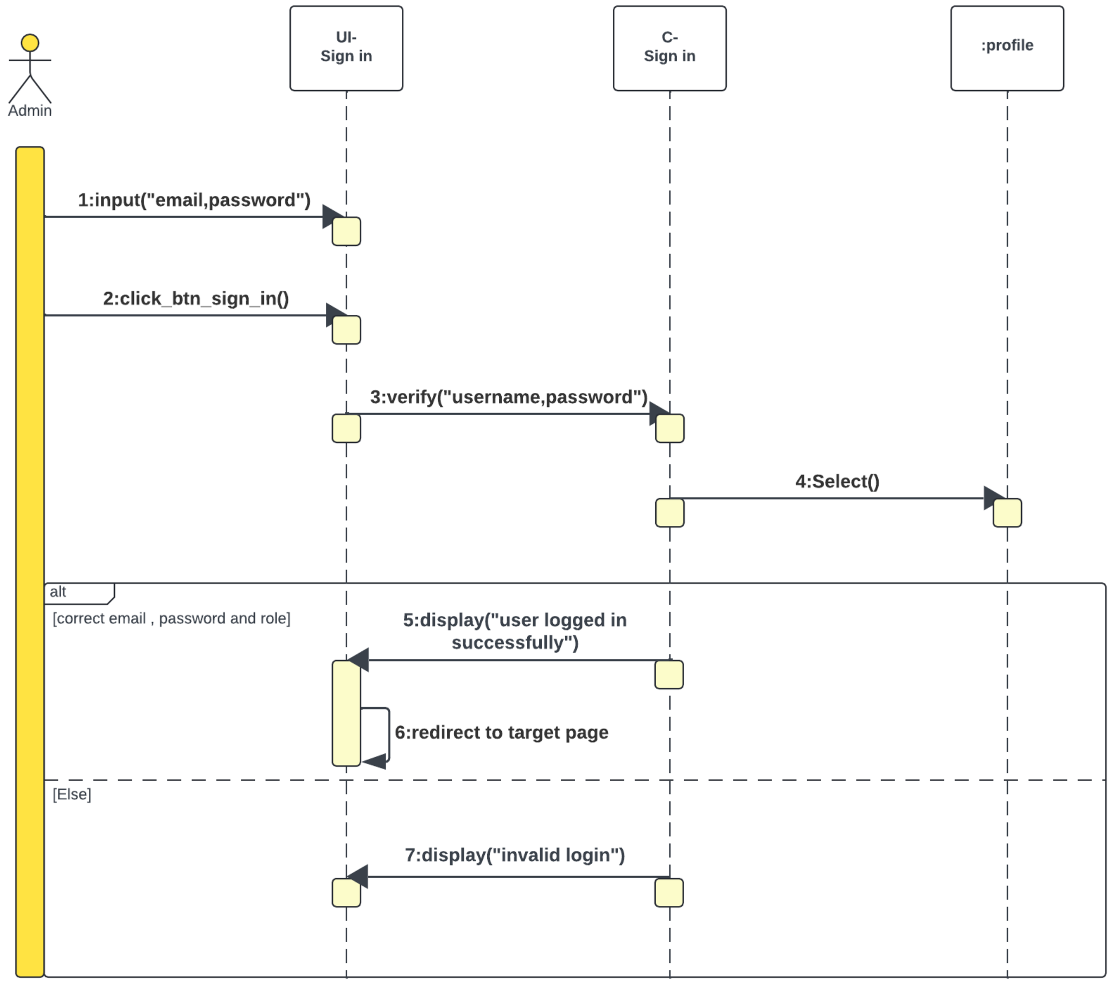

Figure 3.4: Sequence Diagram "Authenticate"

3.4.2 Conception of the Use Case "View Information"

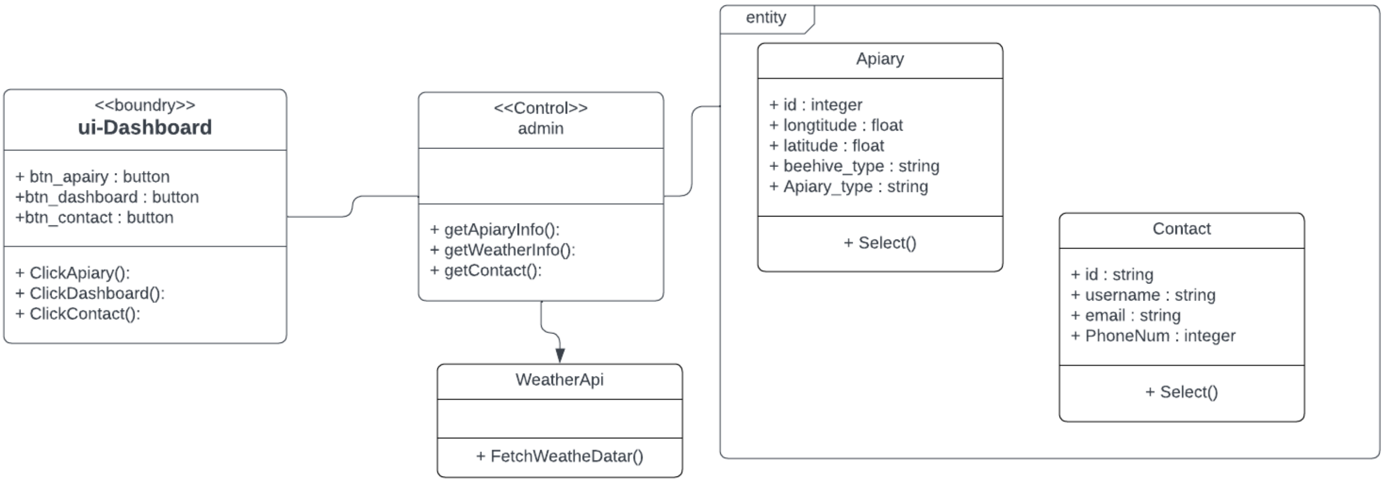

Figure 3.5: Class Diagram "View information"

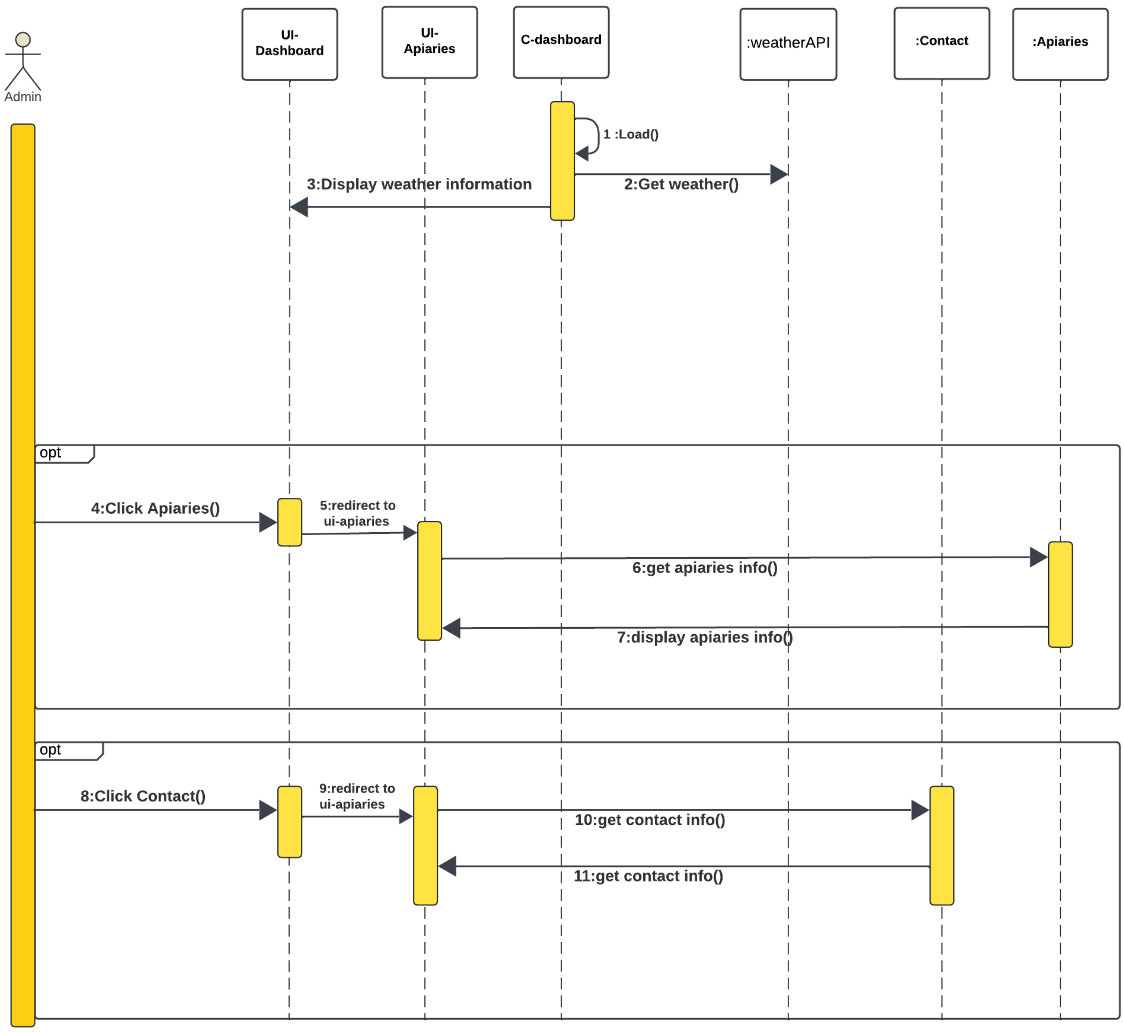

Figure 3.6: Sequence Diagram ”View information”

3.5 Implementation of Sprint 0

3.5.1 Implementation of the Use Case "Authenticate"

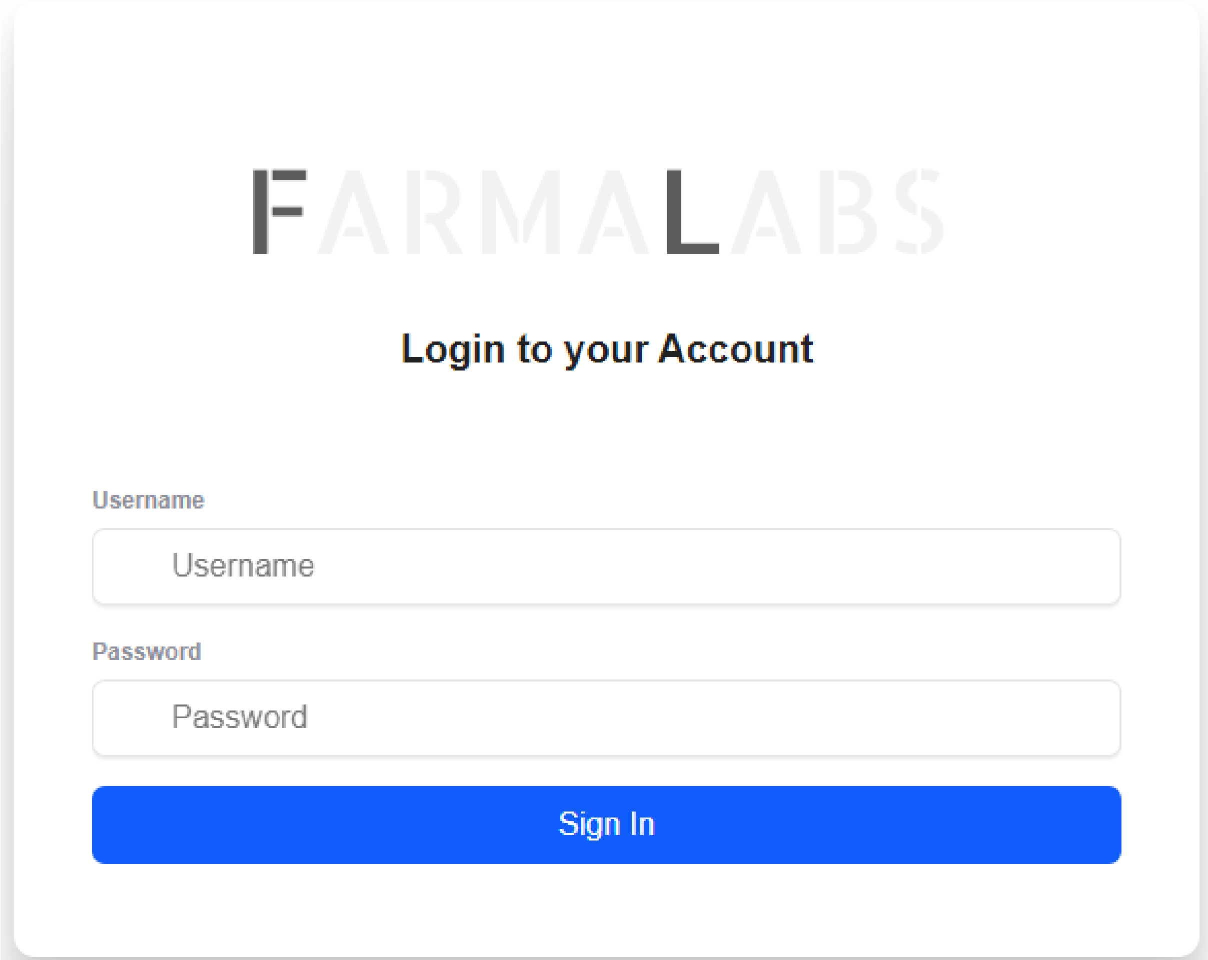

Figure 3.7: interface "Authenticate"

3.5.2 Implementation of the Use Case "View information"

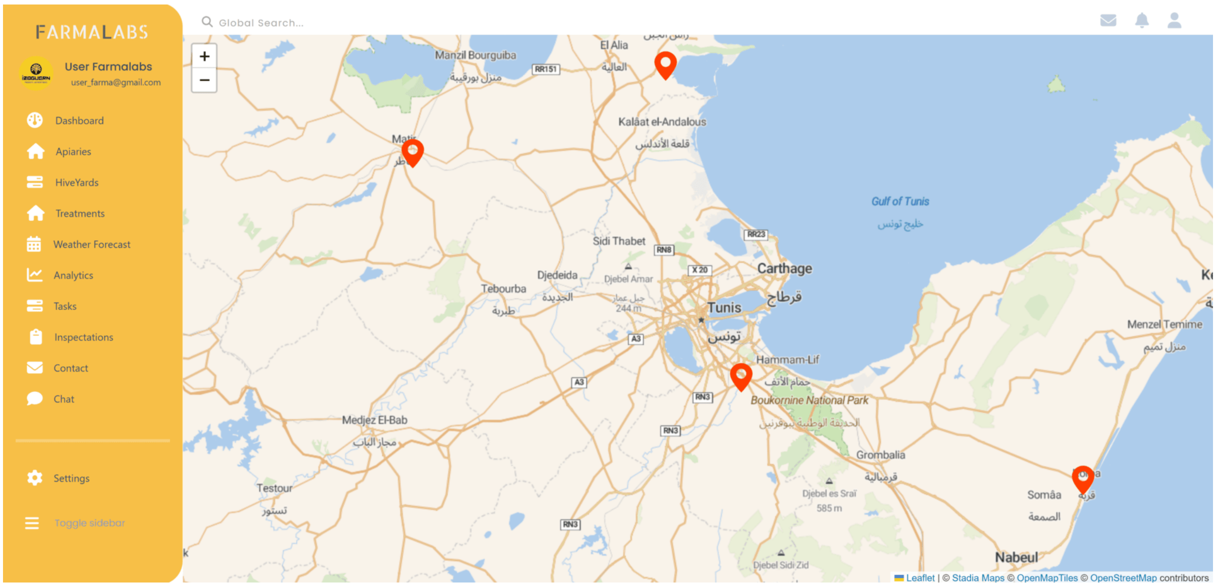

Figure 3.8: interface "View information"

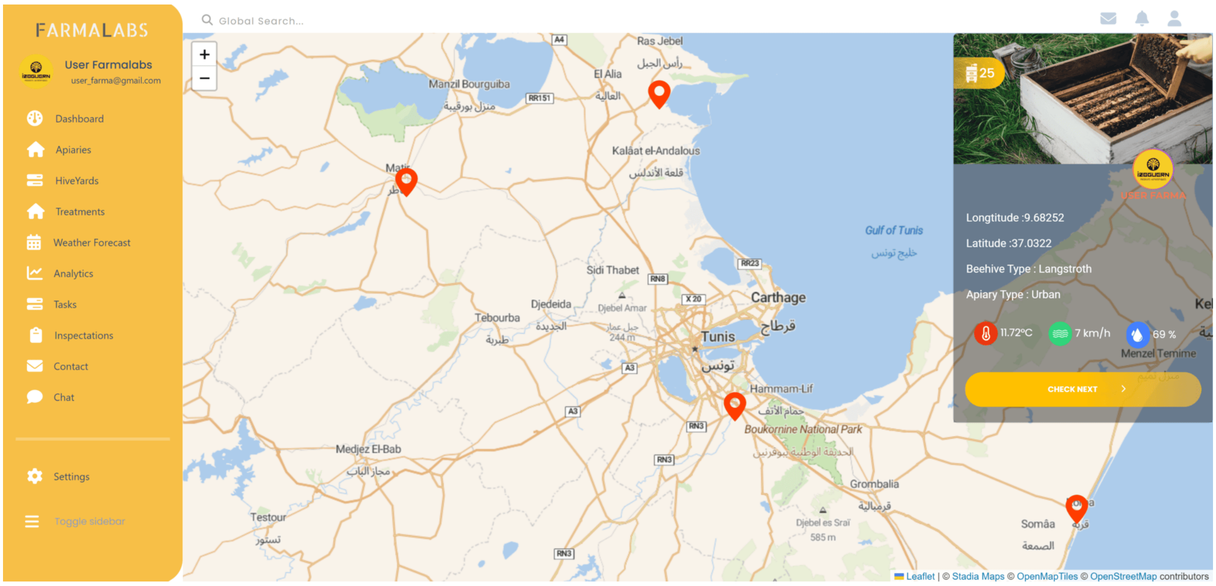

Figure 3.9: interface "View information"

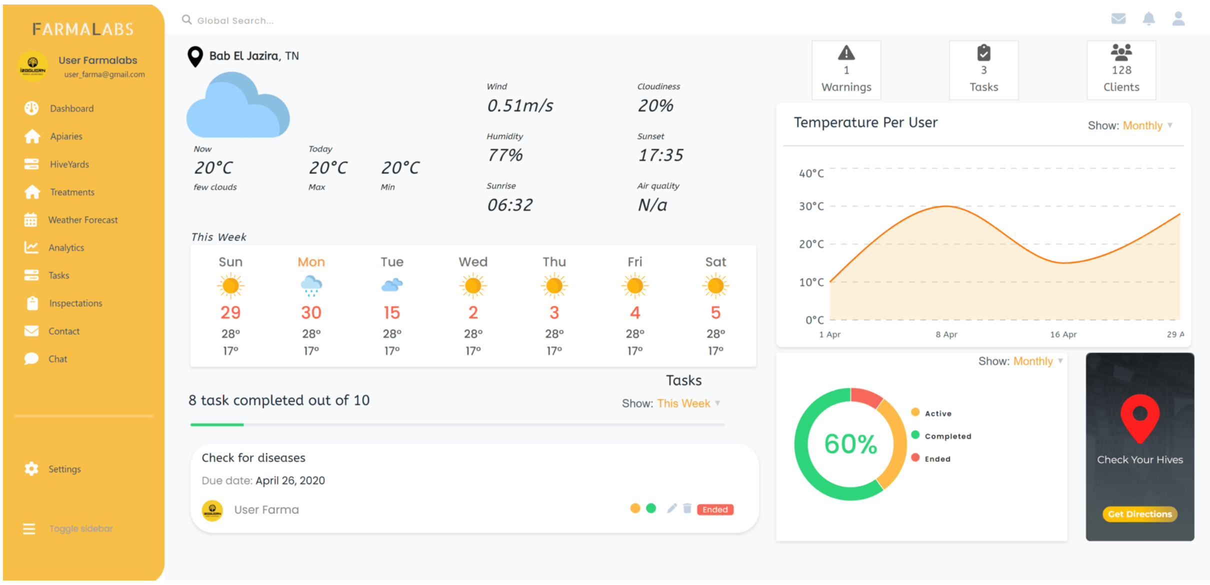

Figure 3.10: interface "View information"

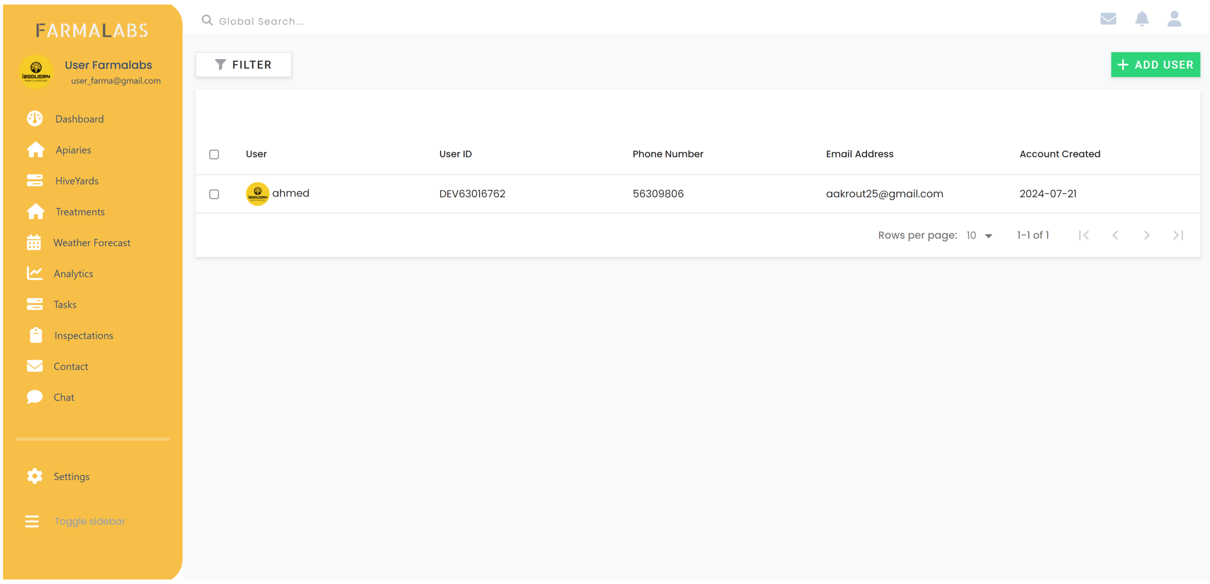

Figure 3.11: interface "View information"

3.6 Conclusion

Sprint 0 established the foundation for the Farmalabs project, covering the key functionality of user authentication and viewing information. The next sprints will build on this foundation to further develop the application.

Chapter 4

Sprint 1

4.1 Introduction

In this chapter, Sprint 1 focuses on developing key features for the Farmalabs admin dashboard. The primary objectives are to enable the Admin to efficiently manage treatments, tasks, and inspections. This sprint aims to ensure smooth dashboard operations while prioritizing data accuracy and user experience, supporting the ongoing management of apiary activities.

4.2 Identification of Sprint 1 Backlog

Product Backlog	**Priority**	**Estimation**	**Sprint**
As an Admin, i can manage treatments	2	Medium	sprint 1
As an Admin, i can manage tasks	2	Medium	sprint 1
As an Admin, i can manage inspectation	2	Medium	sprint 1

Table 4.1: Sprint 1 backlog

4.3 Sprint 1 Refinement

In this section, we explore the following use cases:

- Manage Treatment
- Manage Task
- Manage Inspectation

4.3.1 Refinement of the use case "Manage Treatment"

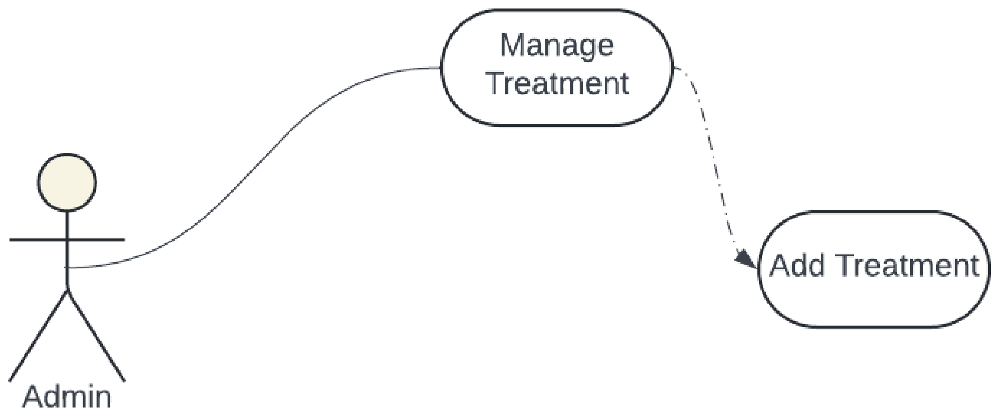

Figure 4.1: Use Case "Manage Treatment"

Use case	– Authenticate
Actor	– Admin
Pre-Conditions	– System running – Internet connection – Access to the application
Post-Conditions	– View Treatments/ schedule Treatment
Main Scenario	– The Admin selects treatment page . – The system displays the list of treatments – The Admin chooses to schedule a new treatment – The Admin inputs the treatment details and schedule time – The Admin presses the Create button – The system confirms the scheduling of the treatment

Table 4.2: Manage Treatment refinement table

4.3.2 Refinement of the use case "Manage Task"

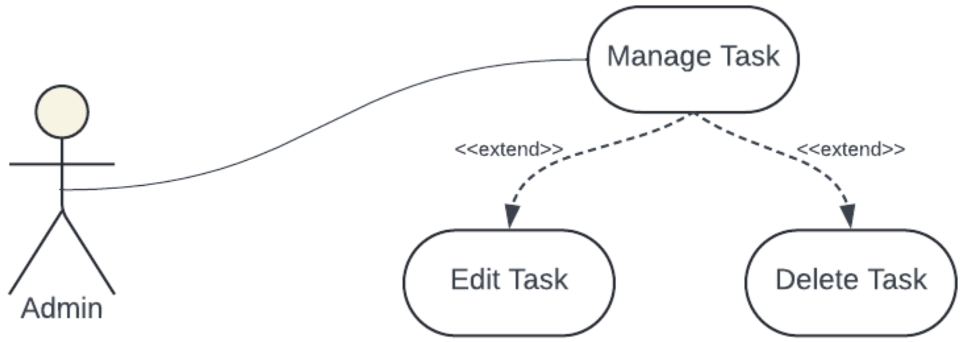

Figure 4.2: Use Case "Manage Task"

Use case	– Manage Task
Actor	– Admin
Pre-Conditions	– System running – Internet connection – Access to the application
Post-Conditions	– Task edited/deleted
Main Scenario	– The Admin selects the task management page. – The system displays the list of tasks – The Admin chooses to edit or delete an existing task – The Admin updates task details or confirms task deletion – The Admin presses the Save/Delete button – The system confirms the task has been updated/deleted

Table 4.3: Manage Task refinement table

4.3.3 Refinement of the use case "Manage Inspectation"

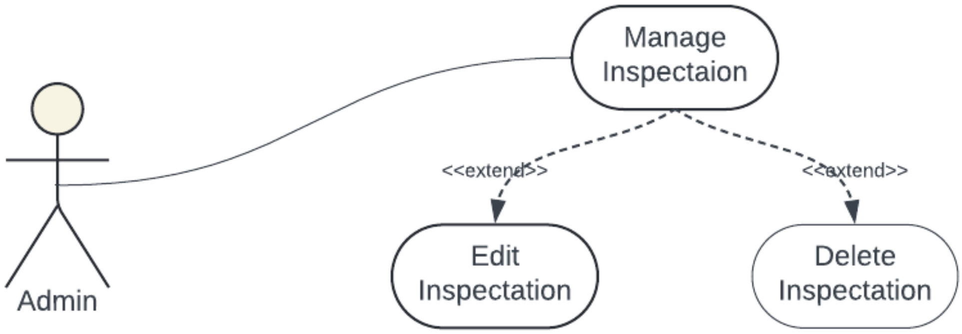

Figure 4.3: Use Case "Manage Task"

Use case	– Edit/Delete Inspection
Actor	– Admin
Pre-Conditions	– System running – Internet connection – Access to the application
Post-Conditions	– Inspection details updated or deleted
Main Scenario	– The Admin selects the inspection management page. – The system displays the list of inspections – The Admin chooses to edit or delete an inspection – The Admin updates the inspection details or confirms deletion – The Admin presses the Save/Delete button – The system confirms the inspection has been updated or deleted

Table 4.4: Manage Inspection refinement table

4.4 Sprint 1 Conception

4.4.1 conception of the use case "Manage Treatment"

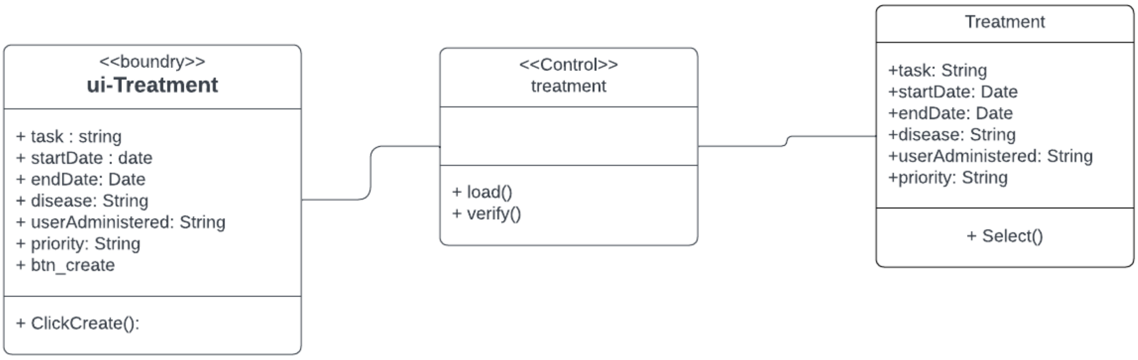

Figure 4.4: Class Diagram "Manage Treatment"

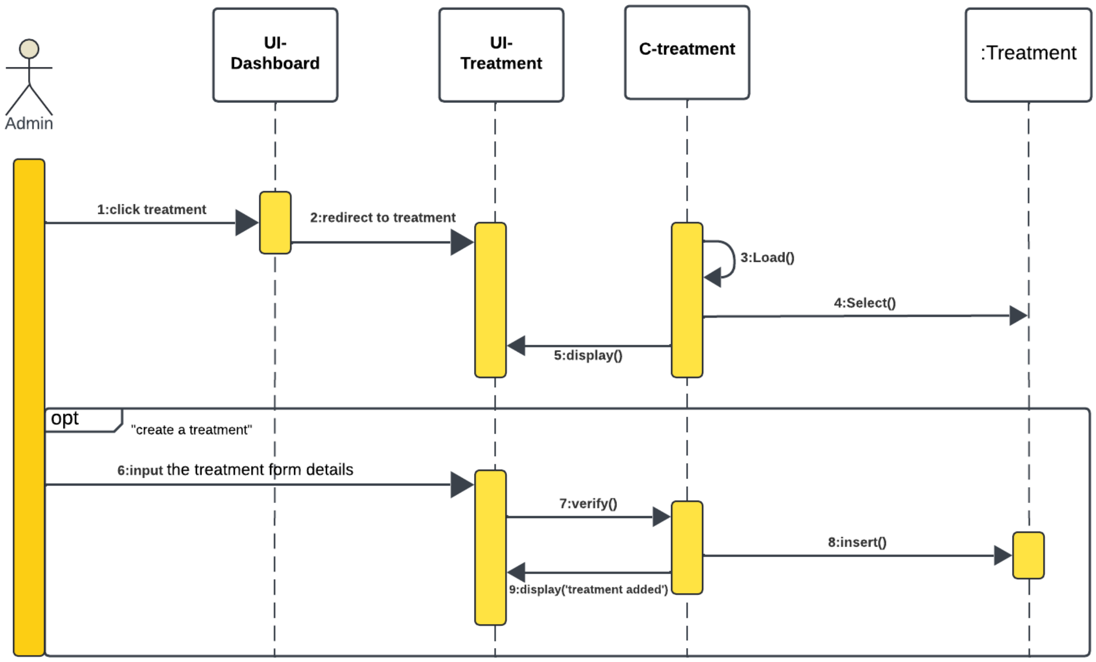

Figure 4.5: Sequence Diagram "Manage Treatment"

4.4.2 conception of the use case ”Manage Task”

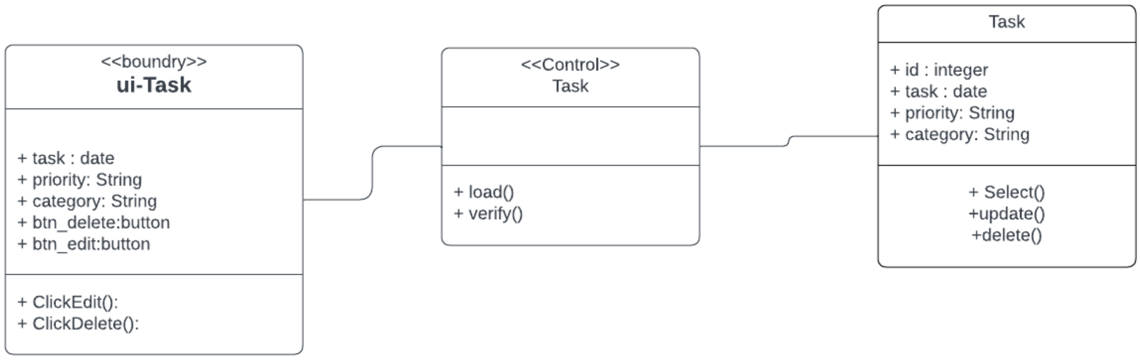

Figure 4.6: Class Diagram ”Manage Task”

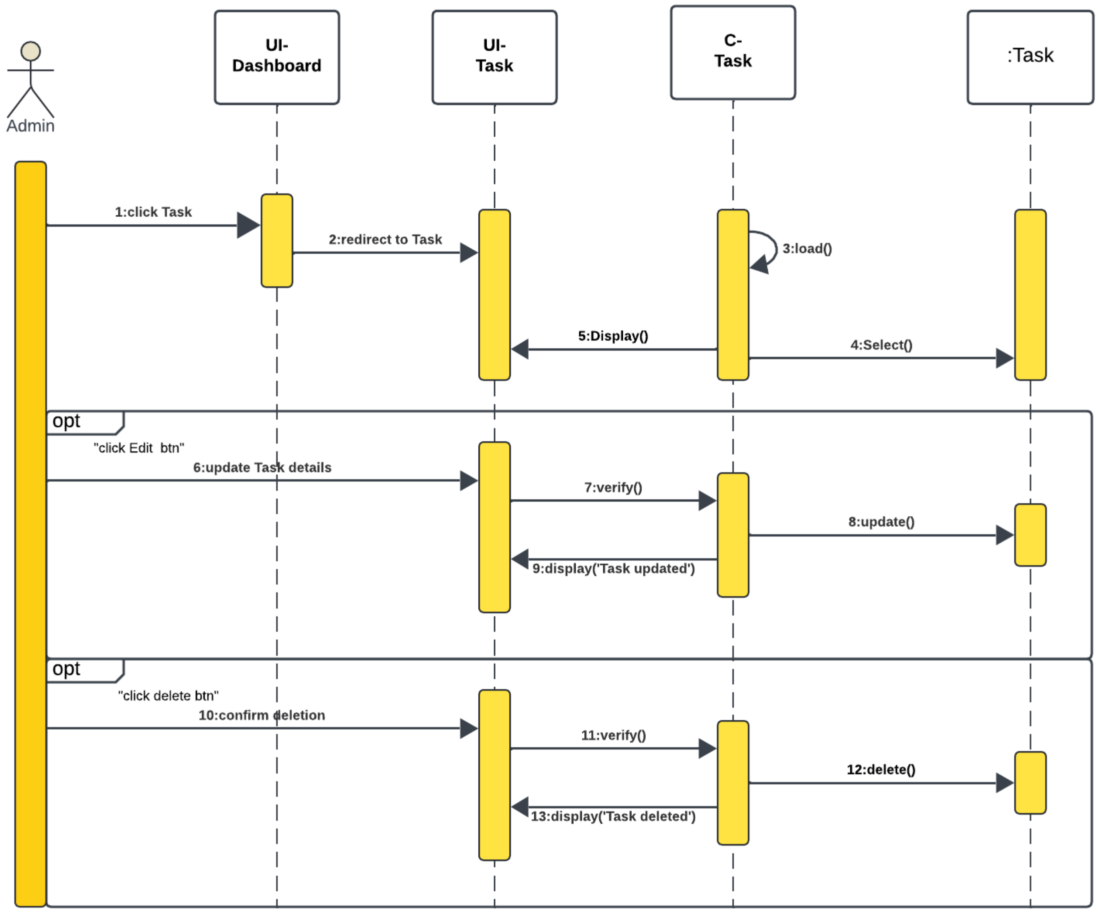

Figure 4.7: Sequence Diagram ”Manage Task”

4.4.3 conception of the use case "Manage Inspectation"

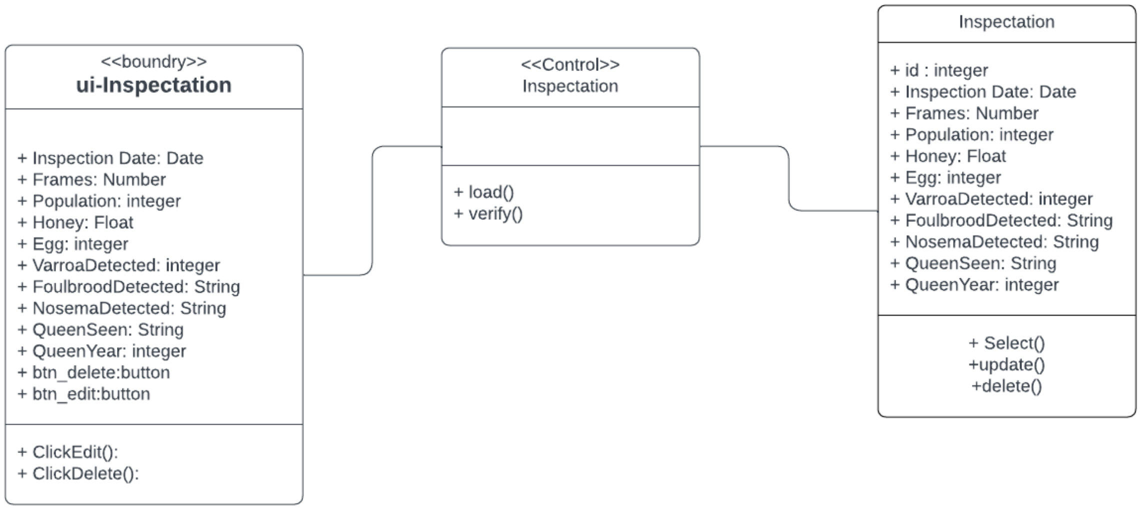

Figure 4.8: Class Diagram "Manage Inspectation"

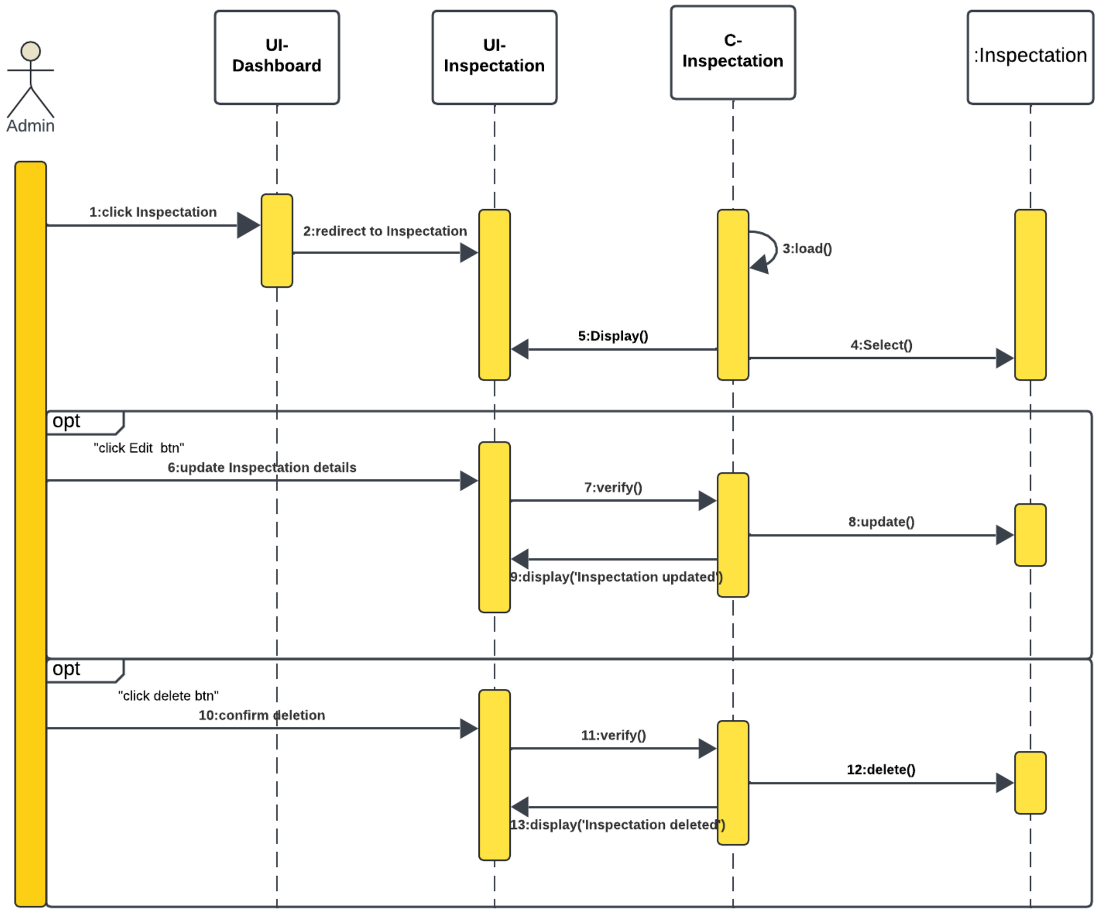

Figure 4.9: Sequence Diagram "Manage Inspectation"

4.4.4 Implementation of the Use Case "Manage Treatment"

The Treatment management interface allows the user to Add custom treatment .

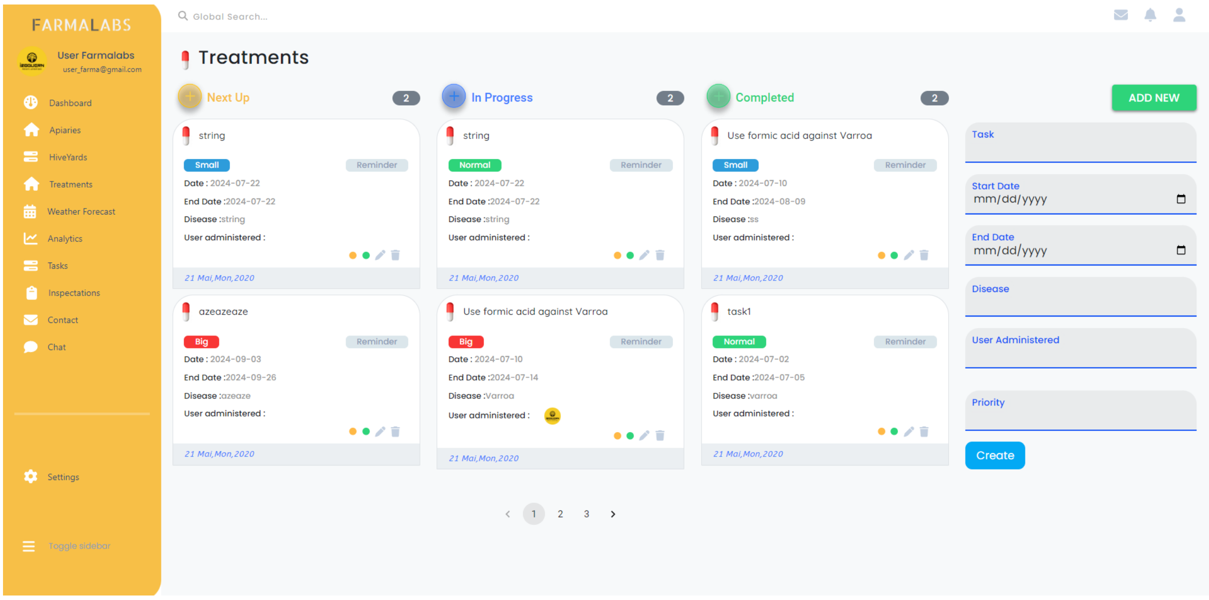

Figure 4.10: interface "Manage Treatment"

4.4.5 Implementation of the Use Case "Manage Task"

The task management interface allows the user to edit and delete Tasks.

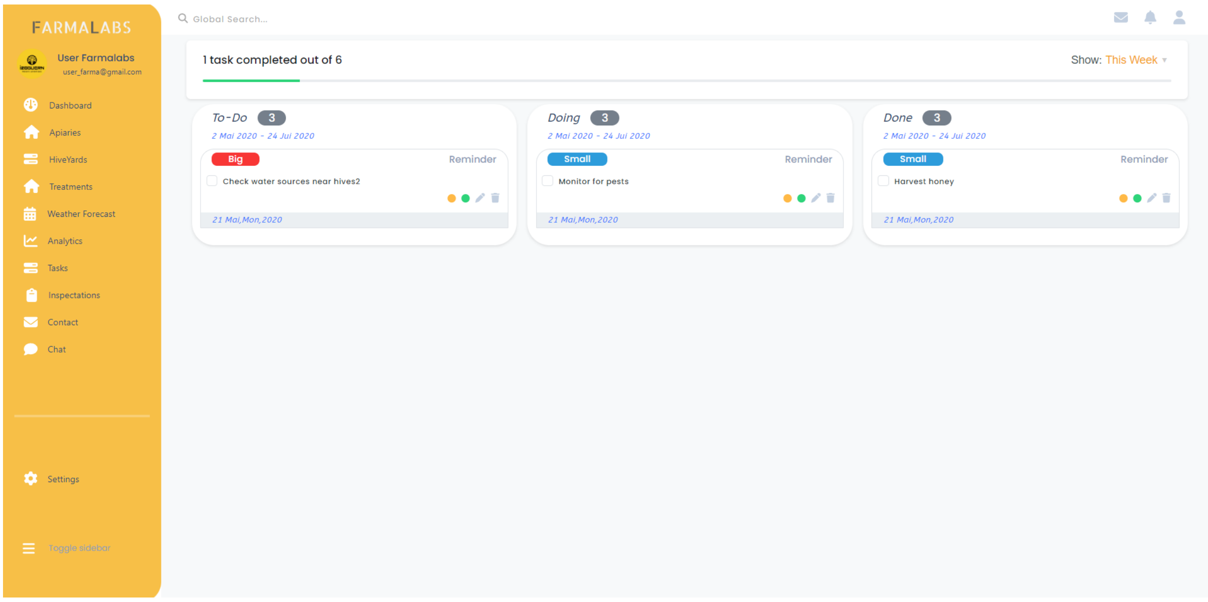

Figure 4.11: interface "Manage Task"

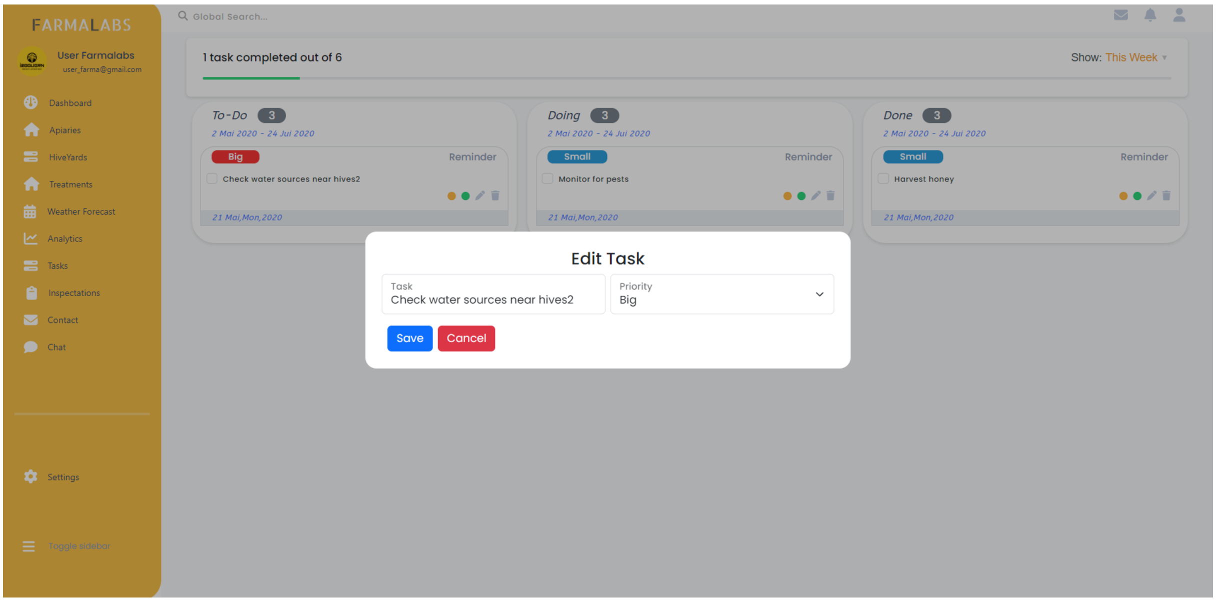

Figure 4.12: interface "Manage Task"

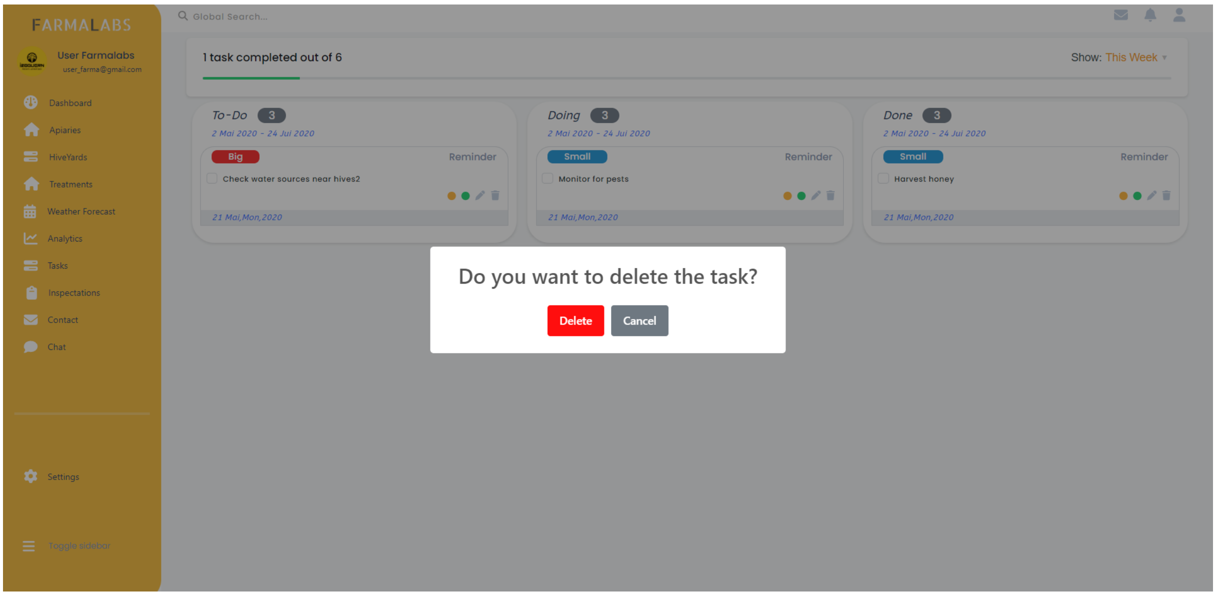

Figure 4.13: interface "Manage Task"

4.4.6 Implementation of the Use Case "Manage Inspectation"

The Inscpectation management interface allows the user to edit and delete inspectation.

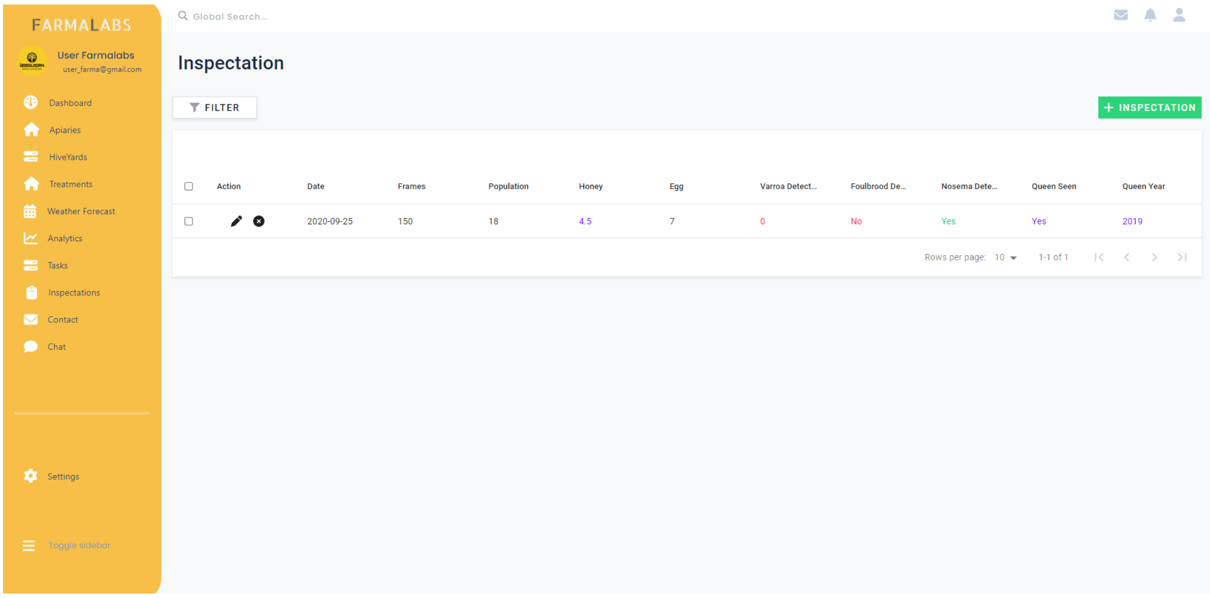

Figure 4.14: interface "Manage Inspectation"

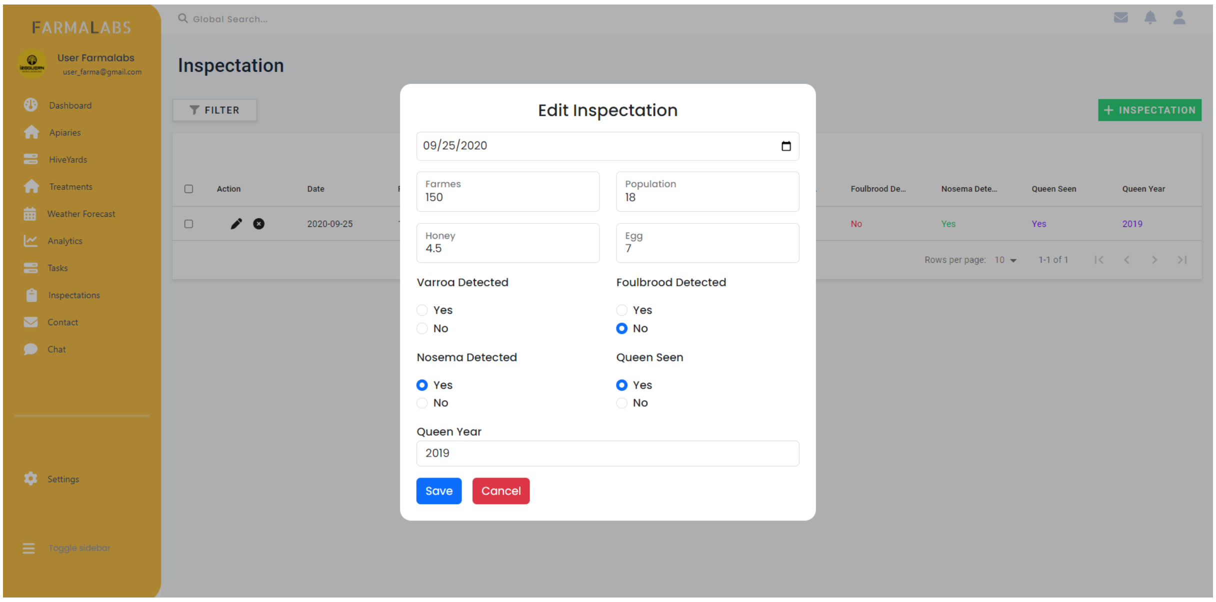

Figure 4.15: interface "Manage Inspectation"

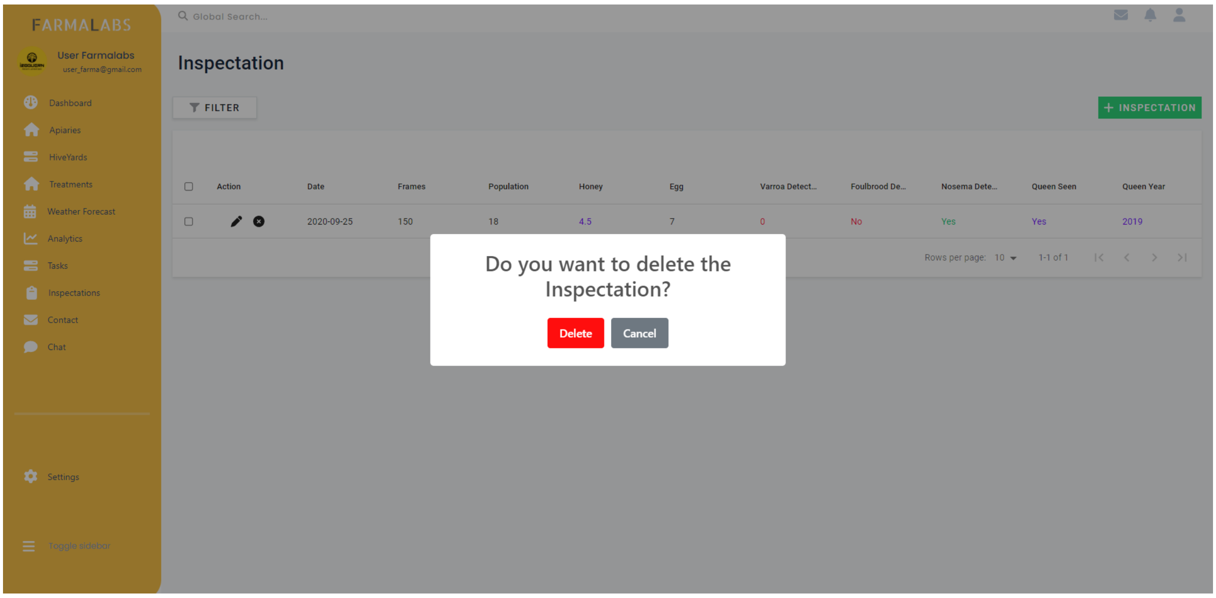

Figure 4.16: interface "Manage Inspectation"

4.5 Conclution

Sprint 1 has focused on improving the Farmalabs admin dashboard by enabling effective management of treatments, tasks, and inspections. This sprint has prioritized user experience ensuring smooth operations for managing apiary activities.

General Conclusion

In conclusion, The Farmalabs platform provides a simple and efficient solution for managing apiaries. It integrates key features such as treatment scheduling, task management, inspections, and real-time weather updates, making it easier for beekeepers to oversee their operations. Built with modern web technologies, the system is responsive, scalable, and user-friendly, ensuring a smooth experience for administrators. This report has outlined the system's major functionalities, design approach, and implementation, highlighting how Farmalabs helps optimize the management of apiaries, improves daily workflows, and supports better decision-making for administrators.

Webography

[1] Lucidchart: https://www.designproject.io/blog/creating-better-diagrams-with-lucidchart/

[2] ReactJs: https://www.geeksforgeeks.org/reactjs-introduction/

[3] FastApi: https://fastapi.tiangolo.com/

[4] Bootstrap: https://wpdean.com/what-is-bootstrap/

[5] CSS3: https://www.simplilearn.com/difference-between-css-and-css3-article

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[8] VSCode: https://code.visualstudio.com/docs/supporting/FAQ